

KUTCHAN, Japan

WMO: 474330

Lat: 42.9011N

Lon: 140.7581E

Elev: 617

StdP: 14.37

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	8.6	1.0	5.9	5.5	4.9	7.2	9.7	26.2	30.2	22.4	26.8	2.9	50	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	82.4	72.1	79.6	70.7	77.2	69.2	74.3	79.8	72.8	77.5	71.3	75.6	10.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	123.3	77.6	71.2	117.6	76.2	69.7	111.8	74.7	38.3	79.9	37.0	77.6	35.7	75.9	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	18.7	16.3	-3.7	88.0	4.1	2.7	-6.7	90.0	-9.1	91.6	-11.4	93.1	-14.4	95.1
			-4.2	76.3	4.1	2.0	-7.1	77.7	-9.5	78.9	-11.8	80.1	-14.8	81.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	21.3	22.6	29.5	40.7	52.4	60.7	68.3	69.9	61.8	49.4	37.0	25.9
	DBStd	18.08	5.84	6.10	6.18	6.22	5.71	5.16	4.50	4.64	5.89	5.99	7.20	6.32
	HDD50	3847	888	767	637	288	39	1	0	0	1	85	393	748
	HDD65	7607	1353	1187	1102	729	390	147	19	11	131	485	840	1213
	CDD50	2052	0	0	0	9	115	321	567	617	354	66	3	0
	CDD65	336	0	0	0	0	1	17	121	163	34	0	0	0
	CDH74	1398	0	0	0	0	28	95	429	713	133	0	0	0
	CDH80	212	0	0	0	0	5	10	60	123	14	0	0	0
Wind	WSAvg	6.9	6.6	7.0	7.5	7.3	7.9	7.4	6.9	6.1	5.8	6.3	6.8	6.8
Precipitation	PrecAvg	58.5	5.5	4.3	3.5	3.3	3.5	4.0	6.1	5.8	6.2	4.5	5.6	6.2
	PrecMax	68.5	8.4	7.4	5.7	5.4	5.2	6.3	10.6	11.2	11.9	6.8	10.0	11.3
	PrecMin	46.6	3.2	1.7	1.5	1.4	1.9	2.4	3.8	2.5	2.7	1.8	3.1	3.1
	PrecStd	5.9	1.3	1.3	1.2	1.1	1.0	0.9	1.8	2.2	2.5	1.3	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	42.6	49.6	67.3	77.6	80.9	85.2	87.2	81.8	68.9	60.1	47.2
		MCWB	36.0	38.2	41.5	53.0	60.8	66.4	74.0	75.8	71.6	59.9	53.1	43.5
	2%	DB	35.6	38.5	45.6	60.9	71.6	76.6	81.4	83.3	77.6	65.7	55.8	41.7
		MCWB	33.1	34.7	39.9	50.2	58.3	64.3	71.9	73.6	68.7	58.4	50.5	38.8
	5%	DB	33.2	35.6	42.6	55.7	67.5	73.4	78.5	80.4	74.6	63.1	52.5	38.5
		MCWB	31.0	32.5	37.9	47.0	55.2	62.9	70.7	72.1	67.1	56.8	48.1	35.7
	10%	DB	31.0	33.2	39.9	51.7	63.6	70.4	76.0	77.9	72.1	60.6	49.5	35.6
		MCWB	28.8	30.5	35.7	44.6	53.7	62.1	69.3	70.7	65.7	54.9	45.6	33.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	39.2	43.3	55.1	62.0	68.9	75.6	77.1	73.7	62.5	54.7	43.8
		MCDB	38.1	42.1	47.2	64.0	72.8	76.5	82.3	85.0	80.0	65.5	58.6	46.4
	2%	WB	33.6	35.3	40.8	51.1	59.3	66.5	73.6	75.0	71.3	60.1	51.2	39.3
		MCDB	35.3	37.9	44.8	59.3	69.0	73.2	78.7	80.9	75.5	63.8	54.8	41.5
	5%	WB	31.4	33.1	38.4	48.1	57.2	64.7	72.2	73.5	69.1	57.9	48.5	36.0
		MCDB	33.1	35.4	42.1	54.7	64.7	71.2	76.8	78.4	72.8	62.0	52.1	38.1
	10%	WB	28.8	30.6	36.0	45.2	55.2	63.0	70.7	72.2	66.6	55.6	45.7	33.3
		MCDB	30.7	33.1	39.5	50.8	61.9	68.9	74.7	76.6	71.0	59.9	49.3	35.3
Mean Daily Temperature Range	5% DB	MDBR	12.4	13.8	14.3	16.0	17.8	15.8	13.5	13.9	17.3	17.5	13.1	11.2
		MCDBR	14.9	17.1	17.7	23.3	25.9	21.6	17.2	16.9	18.9	20.8	19.5	15.3
		MCWBR	13.2	14.4	13.1	13.8	13.8	11.1	8.8	8.7	10.8	14.1	14.9	13.2
	5% WB	MCDBR	14.3	16.5	16.6	21.0	21.8	17.7	14.0	14.0	15.1	17.9	18.5	14.7
		MCWBR	13.2	14.4	13.1	13.6	12.4	9.7	7.6	7.8	9.5	13.1	15.0	13.3
Clear-Sky Solar Irradiance	taub	0.342	0.392	0.459	0.519	0.535	0.514	0.498	0.465	0.410	0.402	0.386	0.361	
	taud	2.019	1.879	1.791	1.787	1.846	1.980	2.097	2.211	2.314	2.191	2.128	2.022	
	Ebn at Noon	237	243	242	239	239	246	248	252	259	241	221	215	
	Edh at Noon	36	49	61	66	64	56	49	43	36	36	32	33	
All-Sky Solar Radiation	RadAvg	384	612	1025	1421	1603	1549	1385	1355	1176	839	468	319	
	RadStd	27	57	84	147	149	150	147	109	116	50	48	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page